A Capable, Fully Tensor-Decomposable Vision-Language-Action Policy

Exact rational conversion, protocol-aligned control, and weight-derived analysis

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Abstract

Tensor decomposition factors a large table of interactions into a small set of low-rank components that can be ranked and inspected. Applying it directly to a conventional robot policy is difficult because softmax, ordinary activations, and normalization interrupt the required algebraic form. We introduce χ-VLA, a compact vision-language-action policy whose learned map uses only bilinear, linear, and rational operations. Its checkpoint therefore defines an explicit rational tensor program, making interactions among image, instruction, state, and action coordinates accessible from the weights. A runtime operator audit of the seed-0 20.1M ViT checkpoint finds no incompatible learned operation, and local reconstructions reproduce its deployed rational and bilinear branches near the float32 precision floor. Across three independently trained 20.1M ViT policies, closed-loop LIBERO-Object success is $85.3\% \pm 4.6\%$ over 500 canonical trials per checkpoint. A preregistered elementwise prediction-mean ensemble reaches $467/500 = 93.4\%$ at the cost of three forward passes. On a separate three-checkpoint convolutional instantiation, independent reconstruction covers all $4{,}608$ joint-attention head-input cases with worst relative error 6.36×10^{-16}. Exact-attention records naming the same three ViT checkpoint artifacts derive the softmax-free attention decomposition and numerically replay every attention module on one fixed cached input per checkpoint, with worst relative error 2.56×10^{-7}. In a seed-0 ViT checkpoint, all 36 primary block-head tests favor its rank-128 subspace over equal-rank random controls, with median fidelity ratio 2.44. Separately, prospectively fixed rank-96 visual projectors retain 95.2–98.2% of baseline successes on tasks absent from corrected projector construction, while three equal-rank random projectors retain only 3.3–13.6%. The evidence establishes strong specialist capability and exact layerwise access to trained attention coefficients. It does not establish broad robotic generalization, robust instruction grounding, or an exact end-to-end decomposition of the full policy.

1 Introduction

Modern robot policies can work well while remaining difficult to inspect. A checkpoint contains millions of learned numbers, but it does not directly reveal which combinations of image regions, instruction tokens, and robot state produce an action. Analysts therefore usually collect activations on a dataset and fit a separate probe. That can identify correlations, but it leaves open whether the same structure is present in the weights and whether it causes behavior.

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Tensor decomposition offers a different analysis surface. A tensor is a multidimensional table, and a decomposition rewrites that table as a sum of simpler low-rank factors, much as a matrix factorization finds dominant directions in a matrix. If a network is built from additions, multiplications, and ratios of polynomials, its weights define an explicit table of input interactions. Decomposing that table can rank the combinations the model is constructed to use. By *fully tensor-decomposable*, we mean that every learned block preserves this algebraic form. This is an operator-closure claim, not a claim that we materialize one compact end-to-end coefficient tensor. Its degree and representation size grow rapidly with depth. For example, one coefficient can connect an image coordinate, an instruction coordinate, and an action coordinate. Factoring the coefficient table ranks high-energy cross-modal interactions directly from the checkpoint without first fitting an activation probe. Those factors are candidate structure, not automatically human-readable concepts or causal explanations. The aim is not merely compression. It is a direct path from parameters to candidate mechanisms that can be tested by intervention [3, 4].

Robot control makes this goal useful and difficult. Visual shortcuts can produce high benchmark success while ignoring the instruction, and small action errors compound in closed loop. Standard softmax, nonlinear activations, and square-root normalization break the explicit polynomial form. Continuous actions also cannot rely on the external categorical sampling used by a language model. A practical design must recover these functions without losing precise control.

We study the narrower but testable question:

> *Can a specialist VLA retain strong closed-loop capability when every deployed learned operation is polynomial or rational, while supporting exact weight analysis?*

We test capability and structural analysis in two fully rational encoder instantiations. The same three ViT checkpoints establish closed-loop capability and support the exact attention audit. A separate convolutional instantiation tests whether the structural certificates transfer across visual encoders.

Our contributions are:

1. We construct a complete VLA from tensor-compatible primitives. A rational normalization supplies input-specific scaling without leaving the rational function class.
2. We evaluate three fully rational 20.1M ViT policies on $1{,}500$ canonical LIBERO-Object trials. They reach $85.3\% \pm 4.6\%$, while a fixed prediction-mean ensemble reaches $467/500 = 93.4\%$.
3. We mechanically audit the seed-0 20.1M ViT checkpoint and explicitly reconstruct deployed rational and bilinear branches from their weights.
4. We extend exact weight-derived decomposition through individual attention layers with a reduced-Gram calculation that avoids materializing the full high-order tensor. In a separate closed-loop audit, learned rank-96 visual projectors retain 95.2–98.2% of baseline successes, versus 3.3–13.6% for random projectors.

This paper does not claim current state of the art. It also does not claim that comparable mechanisms are impossible to obtain from conventional policies. The empirical question is whether the algebraic policy makes exact weight structure directly accessible.

2 A rational tensor robot policy

2.1 Tensor-compatible primitives

Each primitive below replaces a familiar transformer component while keeping an explicit coefficient description. Linear maps contribute first-order terms. Multiplying learned linear projections creates higher-order interactions whose coefficients remain functions of the weights, rather than an unstructured nonlinear black box.

Homogeneous coordinate. We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative layer to represent bias, linear, and quadratic terms in one contraction.

Bilinear feed-forward block. For $L, R \in \mathbb{R}^{r\times(d+1)}$ and $D \in \mathbb{R}^{d\times r}$,

$$\mathrm{BFFN}(x) = D\left[(L\bar{x}) \odot (R\bar{x})\right], \qquad y_o = \sum_{i,j}\left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j}\right)\bar{x}_i \bar{x}_j. \tag{1}$$

The inner sum is an order-three table indexed by output coordinate and two input coordinates. The three matrices give its CP factorization, the tensor analogue of a low-rank matrix factorization, without materializing every table entry.

Softmax-free bilinear attention. Each head forms two query-key systems and one value stream:

$$Y = C\left[M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top)/s\right] V. \tag{2}$$

Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key, value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations. Unlike softmax attention, every multiplicative interaction can be expanded into coefficients determined by the trained weights.

Rational per-instance normalization. RMS normalization contains an input-dependent square root. Let $m(x) = d^{-1}\sum_{j=1}^{d} x_j^2 + \epsilon$, and let $s_0 > 0$ be the frozen running mean-square scale. We deploy a fixed rational approximation $r(v) = P(v)/Q(v)$ to $v^{-1/2}$ over $v \in [0.1, 10]$:

$$v = \frac{m(x)}{s_0}, \qquad \mathrm{RNorm}(x) = \frac{x}{\sqrt{s_0}}\, r(v). \tag{3}$$

Here *rational* means a ratio of two polynomials. Projective coordinates carry each activation as a numerator-denominator pair (N, D) representing N/D. Linear, bilinear, residual, and normalization operations update this pair. The model therefore keeps input-specific rescaling while remaining exactly representable as a rational tensor network. Training and deployment use r itself. The exactness claims below therefore concern faithful evaluation of the deployed rational computation, not uniform agreement with RMSNorm. Appendix A.3 quantifies that distinction.

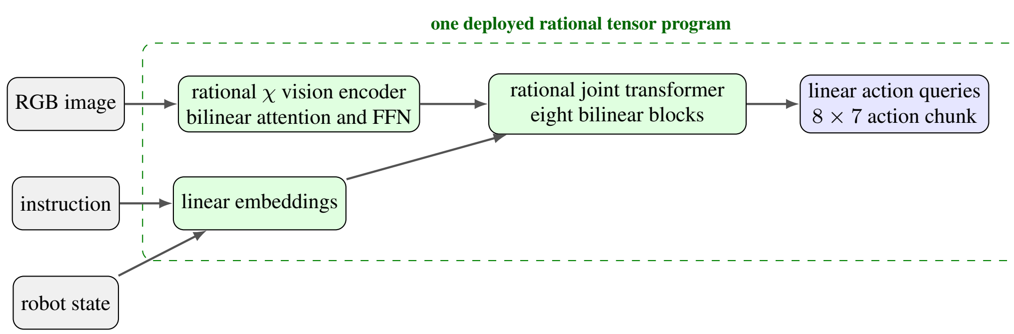

Figure 1: **Deployed χ-VLA map.** The strongest checkpoints use one vision stream and a linear continuous-action head. Every learned component inside the dashed boundary is linear, bilinear, or rational.

2.2 Architecture and training

Both encoder variants follow the high-level VLA sequence of visual encoding, language and state embedding, joint processing, and action decoding [1]. They use a 64×64 agent-view image, an eight-dimensional robot state, a 26-token task vocabulary, and eight action-query tokens. Both share a width-384, eight-block, twelve-head joint backbone and a linear head that maps each action-query state to seven continuous action dimensions. The 20.1M ViT uses four bilinear-attention vision blocks and has 20,137,352 learned parameters. It supplies the closed-loop capability, runtime operator census, exact-attention, and reduced-Gram results. The 19.2M convolutional variant instead uses three stride-2 bilinear convolution stages that produce an 8×8 token grid. Three separate convolutional files receive the joint-attention and rational-normalizer certificates reported in Appendix A.3.

Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay, bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks that verify every requested state was loaded.

2.3 Mechanical certificate on the seed-0 20.1M ViT checkpoint

Architecture definitions can differ from deployed code. We therefore audit the deployed seed-0 20.1M ViT checkpoint at runtime and reconstruct representative operations from saved weights.

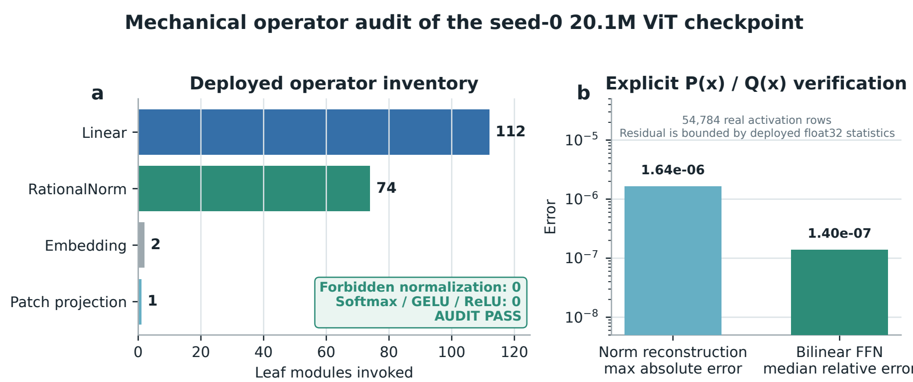

Figure 2: **Mechanical audit of the seed-0 20.1M ViT rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.

The normalization reconstruction has maximum absolute error 1.64×10^{-6}. A bilinear FFN reconstruction has median relative error 1.40×10^{-7}. These residuals are bounded by the deployed module's float32 statistic. The audit certifies operator membership and local reconstruction. It does not materialize one compact polynomial for the full eight-layer transformer.

3 Protocol-aligned closed-loop capability

3.1 Evaluation protocol

We evaluate all ten LIBERO-Object tasks with the following fixed protocol:

- 50 canonical trials per task
- a 280-step cap
- three independently trained ViT checkpoints
- 500 rollouts per checkpoint and 1,500 rollouts in total

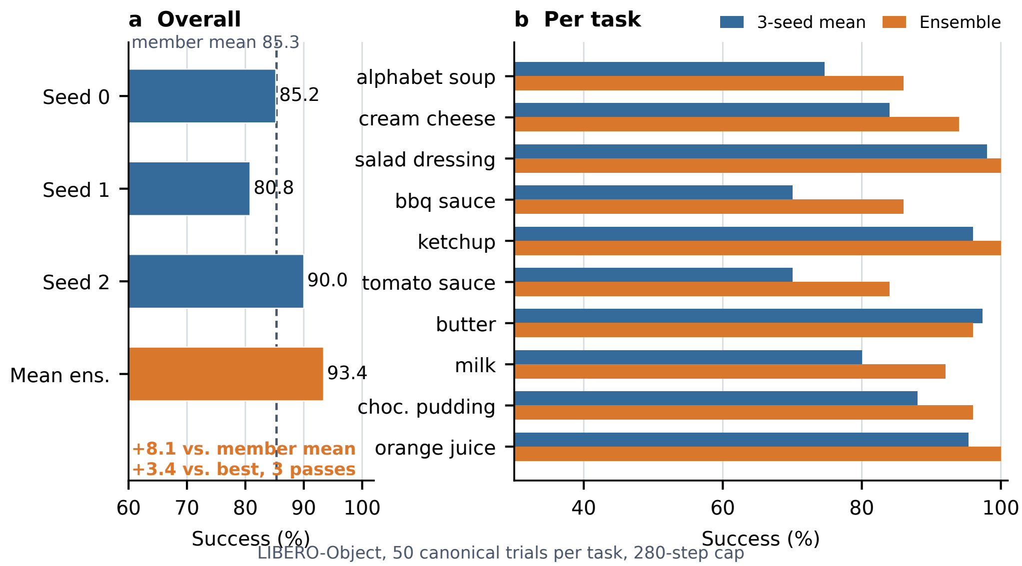

Figure 3: **Verified ViT capability.** Left: three independently trained checkpoints and their fixed elementwise prediction-mean ensemble. Right: per-task three-checkpoint mean and ensemble success. The ensemble uses three forward passes.

3.2 Results

Method	Parameters	Inference	Success
χ-VLA, rational ViT	20.1M	1 pass	$\mathbf{85.3\% \pm 4.6\%}$
χ-VLA, prediction-mean ensemble	$3 \times$ 20.1M	3 passes	$\mathbf{93.4\%}$

Table 1: **Closed-loop LIBERO-Object success.** Each ViT checkpoint receives 500 canonical trials. The single-model row is the mean $\pm$ sample standard deviation across three checkpoints. The ensemble is one fixed composite evaluated on the same 500 states.

The three checkpoints reach $426/500 = 85.2\%$, $404/500 = 80.8\%$, and $450/500 = 90.0\%$. Elementwise averaging of their predicted action chunks reaches $467/500 = 93.4\%$. This is 8.1 points above the member mean and 3.4 points above the strongest member. The ensemble requires three complete forward passes and is not one 20.1M-parameter policy. Every episode-level record, task index, and protocol field is retained. The artifact manifest binds each immutable result file and recorded checkpoint basename to a checkpoint hash verified separately. Member and ensemble shards span different recorded GPU families and predate explicit matrix-precision fields, so the arithmetic gaps are descriptive rather than a matched-hardware ensemble ablation.

3.3 What this result does and does not establish

The result shows that the rational architecture supports high closed-loop specialist capability. It is an existence claim for a compact tensor-decomposable policy, not a claim of broad robotic generalization or state of the art.

4 Exact weight-derived structure through attention

4.1 Exact layerwise construction

Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives a high-order coefficient table. Each entry says how strongly one combination of query, key, and value coordinates contributes to an output. It is *weight-derived* because no examples or fitted probe are needed to construct it. On a tiny four-token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to 2.00×10^{-15}. The internal core identity agrees to 1.24×10^{-14}.

The deployed attention also normalizes each query and key with Equation 3. Each normalization multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled outside the bilinear dot products. The polynomial attention core is unchanged, while explicit numerator and denominator factors carry the normalization. The rational extension agrees to 3.61×10^{-16}. On one fixed cached input per checkpoint, we numerically replay all four vision and eight joint attention modules in each of three records that name the same ViT checkpoint artifacts as the capability results. This covers 36 module-input cases and 384 architectural heads. The worst relative ℓ_2 error is 2.56×10^{-7} and the worst absolute difference is 2.23×10^{-5}. This residual comes from the deployed implementation's intentional float32 statistic, while the reconstruction itself uses the learned coefficients. The algebraic identity is input-general. The fixed inputs provide a numerical replay audit.

We separately certify three convolutional checkpoints from the same rational architecture family. Their vision stacks contain no attention, so we independently rebuild all eight joint modules and all twelve heads per module from the saved weights. Sixteen deterministically selected cache inputs per checkpoint yield 384 module cases and 4,608 head-input cases. Every output is finite and the worst relative ℓ_2 error is 6.36×10^{-16}, far below the frozen 10^{-6} gate. This same-checkpoint certificate covers the joint-attention equation at each normalized layer input. It does not rebuild the convolutional stem, feed-forward blocks, residual composition, final normalization, or action head as one end-to-end contraction.

Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15} values. We instead contract the CP factors into an $O(d^2)$ reduced Gram. This smaller matrix plays the role of a covariance matrix: its leading eigenvectors rank the directions that carry the most coefficient energy without constructing the full table. At toy multi-head scale, it matches brute-force materialization to 2.13×10^{-14}.

4.2 Trained policy result

The reconstruction certificate above spans three ViT checkpoints. The following subspace-ranking experiment is a descriptive analysis of the seed-0 ViT checkpoint. We apply the exact weight-derived calculation to all twelve heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are sampled once, then fixed and reused for all evaluation examples. The weights choose the subspace. Real cached activations are used only afterward to measure how faithfully that subspace preserves the layer's computation.

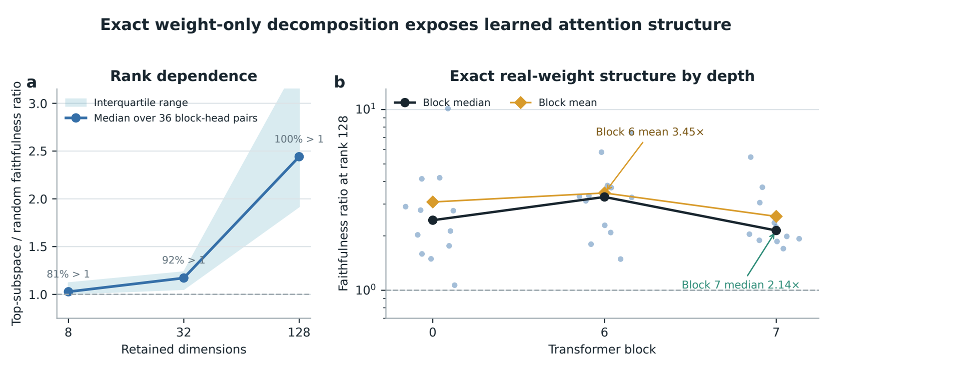

Figure 4: **Exact weight-derived attention structure.** Ratios above one favor the exact weight-derived subspace over a random equal-rank subspace. Each point in the right panel is one trained attention head.

At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks 0, 6, and 7. The 36 block-head pairs are architectural units within one checkpoint, not independent model-level replicates.

This extends exact weight-only construction through individual attention layers. It is not an exact end-to-end decomposition of the complete policy. Composition across all eight blocks still faces rapid degree and representation growth.

A separate intervention tests behavioral sufficiency at the visual-to-joint interface rather than deriving a projector from the exact attention factors. An action-Jacobian projector estimated on official Object tasks 0–3, with rank 96 fixed before corrected outcomes, retains $259/272$ full-policy successes on tasks 8–9 across three ViT checkpoints. These tasks are absent from corrected projector construction. An activation-energy projector retains $267/272$, while three equal-rank random projectors retain $37/272$, $9/272$, and $22/272$. All ten tasks appeared during policy training. This establishes a structured low-dimensional visual bottleneck relative to random projectors, not unseen-task generalization, unique action-Jacobian specificity, or a causal link to the exact weight factors. Appendix A.2 reports the paired controls and stability intervals.

5 Related work

VLA capability. OpenVLA established an open pretrained VLA and standardized LIBERO comparisons [1]. OpenVLA-OFT showed that continuous action chunks and simple regression can substantially improve adaptation and control efficiency [2]. These systems use large pretrained backbones. χ-VLA instead studies a compact from-scratch specialist under an algebraic restriction.

Tensor and polynomial networks. Polynomial networks and tensor networks use explicit multilinear structure for representation and learning [12–14]. Bilinear MLP work showed that weight eigendecomposition can expose low-rank features and circuits [3]. χ-nets introduced ODT for compositional bilinear networks [4]. We extend the setting to a continuous closed-loop policy and give an exact reduced construction for individual softmax-free attention layers.

Causal mechanisms and policy analysis. Bilinear IMPALA policies have combined competitive ProcGen control with weight-derived eigenfilters, followed by activation probing and control [5]. This is the closest architectural precedent for weight-based policy analysis. Recent work steers pretrained VLAs through causally identified activation directions [6]. Sparse feature circuits identify and edit causal feature graphs in conventional language models [7], while nonlinear causal reductions recover behavioral patterns in reinforcement-learning policies from rollout perturbations [8]. These results make a universal separation from conventional policies untenable. Our intended distinction is empirical: exact weight auditing in a strong continuous vision-language policy.

VLA robustness. Recent evaluations show that standard LIBERO success can coexist with weak robustness and language use [9–11]. We therefore treat LIBERO-Object as a bounded specialist capability setting rather than evidence of broad generalization.

6 Limitations and conclusion

χ-VLA demonstrates that a complete rational tensor policy can reach strong specialist closed-loop capability. The capability and exact-attention records name the same three ViT checkpoint artifacts. The deployed seed-0 operator path also passes a mechanical audit.

The scope is deliberately specific. LIBERO-Object is one specialist suite with fixed task instructions, so these results do not establish broad robotic generalization or robust instruction grounding. Exact reconstruction is certified layer by layer and does not yet yield one compact symbolic contraction of the full policy. The three-checkpoint ensemble trades three forward passes for its strongest capability result.

Within that scope, polynomial and rational restrictions are compatible with strong closed-loop control and provide a precise weight-derived analysis surface. The ViT records connect capability, exact attention, and the reduced-Gram screen through common checkpoint basenames. Separate convolutional files replicate the joint-attention equation certificate across another visual encoder. We do not claim a direct coefficient-edit interface.

Reproducibility. Training and evaluation use fixed task lists, explicit camera and action conventions, canonical-state manifests, checkpointed moving-average weights, and per-run result files. Historical Modal cache-backed analyses read dataset-native task metadata. The replacement Athena pipeline pins that metadata and maps dataset identifiers to official identifiers before defining task splits. A source-level audit compared all 271,996 cached frames across Object, Spatial, Goal, and LIBERO-10 with their pinned LeRobot revisions. Task identifiers, language joins, states, actions, and resized images matched exactly for every frame. The anonymous artifact releases raw structural-certificate JSONs and all 2,000 episode rows behind the ViT member and ensemble results. Immutable SHA-256 manifests and standard-library verifiers independently recompute the capability totals, visual-bottleneck counts and intervals, and the convolutional-attention and RationalNorm certificate decisions. Preflight checks fail if canonical initial states cannot be loaded.

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A Supporting results and decisions

A.1 Expanded attention screen

Within the same seed-0 ViT checkpoint, the corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128, $83/96$ block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean is 2.58, and blocks 2 and 3 provide useful depth controls. At ranks 8 and 32, only $54/96$ and $60/96$ pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider tested rank.

A.2 Prospectively fixed-rank visual bottleneck

The behavioral intervention acts at the width-384 visual-to-joint interface and keeps rank 96. Projector construction uses official Object tasks 0–3. Rank 96 was fixed before any corrected-basis outcome was observed, and the frozen evaluation uses tasks 8–9. Across three checkpoints and 300 canonical episodes, the full policy succeeds on 272 trials. The action-Jacobian projector retains $259/272 = 95.2\%$ of those successes, with a paired checkpoint-task-episode resampling interval of 91.2–98.6%. The activation-energy projector retains $267/272 = 98.2\%$, with an interval of 95.8–100.0%. Three fixed equal-rank random projectors retain $37/272 = 13.6\%$, $9/272 = 3.3\%$, and $22/272 = 8.1\%$.

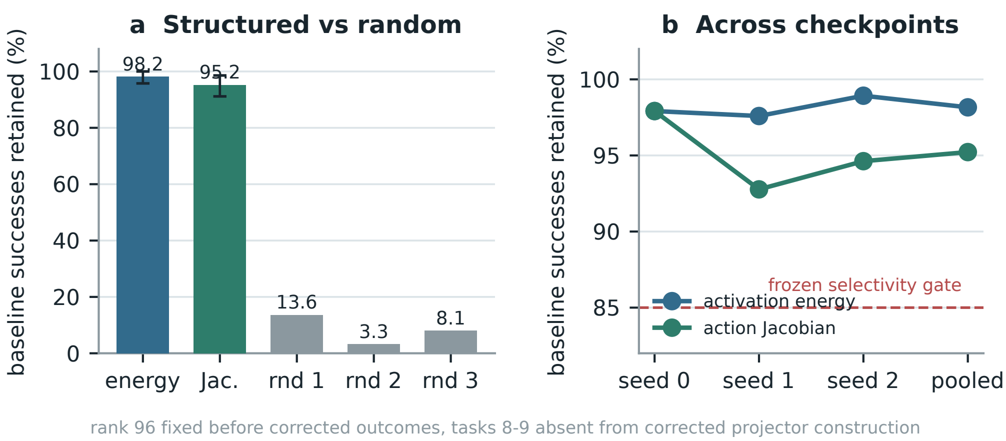

Figure 5: **Corrected closed-loop visual bottleneck.** Both learned rank-96 projectors preserve nearly all baseline successes across three checkpoints, while equal-rank random projectors do not. Error bars in panel (a) are deterministic paired cluster-resampling intervals. Tasks 8–9 are absent from corrected projector construction, but not from policy training. This is not a clean rank-selection-holdout result.

The action-Jacobian basis does not outperform the matched activation-energy basis. The stronger action-specificity gate therefore does not pass. The supported claim is structured visual sufficiency relative to random projectors. This intervention is data-derived and remains separate from the exact weight-only attention decomposition.

A.3 Numerical scope of rational conversion

The deployed degree-$[2/2]$ least-squares rational fit is exact as a rational computation even though it approximates RMS normalization. Its numerator has coefficients $(5.32997, 7.53552, 0.327274)$ and the denominator has coefficients $(1, 9.64827, 2.60664)$. Every denominator coefficient is positive, which certifies $Q(v) > 0$ for all $v \geq 0$. We instrument all 53 RationalNorm module instances, each invoked once per forward pass, on $4{,}096$ deterministic inputs for each of three convolutional checkpoints. Across $544{,}542{,}720$ activation rows, the minimum observed denominator is 1.0016 and no row is nonfinite. Comparing deployed float32 normalization with an otherwise identical pass that recomputes only RationalNorm scales in float64, the largest checkpoint-level primary metric, defined as the maximum of normalized- and physical-action NRMSE, is 2.414×10^{-6} against a frozen 10^{-3} gate.

The certificate also bounds what the operator approximates. Observed normalized mean-square ratios v span 1.66×10^{-4} to 17.17, outside the nominal fit interval $[0.1, 10]$. Across site-checkpoint pairs, the minimum coverages are 60.018% in that interval and 60.022% with rational-versus-exact-RMS scale error at most 3.4%. The frozen 99% secondary coverage gates therefore do not pass. The positive claim is therefore global absence of nonnegative denominator poles and cache-conditioned floating-point fidelity to the deployed rational operator, not uniform agreement with exact RMS scaling. Matched alternative-normalizer runtime overhead remains unmeasured.

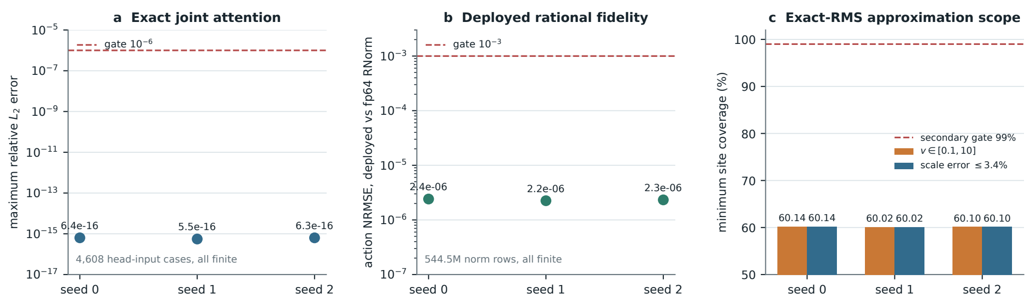

Figure 6: **Frozen structural certificates on three independent convolutional checkpoints.** Panels (a) and (b) pass their primary gates with large margins. Panel (a) reconstructs all $4{,}608$ audited joint-attention head-input cases. Panel (b) compares deployed float32 normalization with an otherwise identical pass that recomputes only RationalNorm scales in float64. Panel (c) reports the separately frozen exact-RMS approximation coverage, which does not pass its 99% secondary gate.